

Zeal College of Engineering and Research

Subject: LP-V

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Class: BE

Div: B

Batch: B1

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Group A: Practical No:3

Program Statement: Implement Min, Max, Sum and Average operations using Parallel Reduction.

Implemented Code:

```
#include <iostream>
#include <omp.h>
using namespace std;

#define MAX 100000
int main() {
    int n;
    cout << "Enter number of elements: ";
    cin >> n;
    int arr[MAX];
    // Input array
    cout << "Enter elements:\n";
    for (int i = 0; i < n; i++)
        cin >> arr[i];
    int min_val = arr[0], max_val = arr[0];
    long long sum = 0;
    double start = omp_get_wtime();

    for (int i = 0; i < n; i++) {
        if (arr[i] < min_val)
            min_val = arr[i];
        if (arr[i] > max_val)
            max_val = arr[i];
        sum += arr[i];
    }
    double end = omp_get_wtime();
    cout << "\n--- Sequential Results ---\n";
    cout << "Min: " << min_val;
    cout << "\nMax: " << max_val;
    cout << "\nSum: " << sum;
    cout << "\nAverage: " << (double)sum / n;
    cout << "\nTime: " << (end - start) << " sec\n";
```

```

int p_min = arr[0], p_max = arr[0];
long long p_sum = 0;
start = omp_get_wtime();
#pragma omp parallel for reduction(+:p_sum) reduction(min:p_min)
reduction(max:p_max)
for (int i = 0; i < n; i++) {
    p_sum += arr[i];

    if (arr[i] < p_min)
        p_min = arr[i];

    if (arr[i] > p_max)
        p_max = arr[i];
}
end = omp_get_wtime();
cout << "\n--- Parallel Results ---\n";
cout << "Min: " << p_min;
cout << "\nMax: " << p_max;
cout << "\nSum: " << p_sum;
cout << "\nAverage: " << (double)p_sum / n;
cout << "\nTime: " << (end - start) << " sec\n";
return 0;
}

```

How to compile: g++ -fopenmp Practical3.cpp -o program

How to run: program.exe

OUTPUT:

```

C:\Windows\System32\cmd.e  X  +  v
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\D_Drive\B.E\Sem-VII\LP-V\HPC>g++ -fopenmp Practical3.cpp -o Practical3

C:\D_Drive\B.E\Sem-VII\LP-V\HPC>Practical3
Enter number of elements: 5
Enter elements:
15
12
90
0
150

--- Sequential Results ---
Min: 0
Max: 150
Sum: 267
Average: 53.4
Time: 0 sec

--- Parallel Results ---
Min: 0
Max: 150
Sum: 267
Average: 53.4
Time: 0.00300002 sec

C:\D_Drive\B.E\Sem-VII\LP-V\HPC>

```